Embeddings for visual analytics

CS424: Visualization & Visual Analytics
Fabio Miranda

urbantk.org

evl electronic visualization laboratory

UNIVERSITY OF ILLINOIS CHICAGO

What are embeddings?

- *Embedding refers to the process and/or the results of projecting data entities into a (relatively) low-dimensional space where the similarities of different points reflect their semantic closeness in the original space.*
- Challenge: learning effective embeddings.

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(Guest Editors)

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Volume 42 (2023), Number 3
STAR – State of The Art Report

VA + Embeddings STAR: A State-of-the-Art Report on the Use of Embeddings in Visual Analytics

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[2]Department of Computer Science and Media Technology, Linnaeus University, Sweden

Abstract
Over the past years, an increasing number of publications in information visualization, especially within the field of visual analytics, have mentioned the term "embedding" when describing the computational approach. Within this context, embeddings are usually (relatively) low-dimensional, distributed representations of various data types (such as texts or graphs), and since they have proven to be extremely useful for a variety of data analysis tasks across various disciplines and fields, they have become widely used. Existing visualization approaches aim to either support exploration and interpretation of the embedding space through visual representation and interaction, or aim to use embeddings as part of the computational pipeline for addressing downstream analytical tasks. To the best of our knowledge, this is the first survey that takes a detailed look at embedding methods through the lens of visual analytics, and the purpose of our survey article is to provide a systematic overview of the state of the art within the emerging field of embedding visualization. We design a categorization scheme for our approach, analyze the current research frontier based on peer-reviewed publications, and discuss existing trends, challenges, and potential research directions for using embeddings in the context of visual analytics. Furthermore, we provide an interactive survey browser for the collected and categorized survey data, which currently includes 122 entries that appeared between 2007 and 2023.

Keywords: embedding techniques, distributed representations, visual analytics, visualization

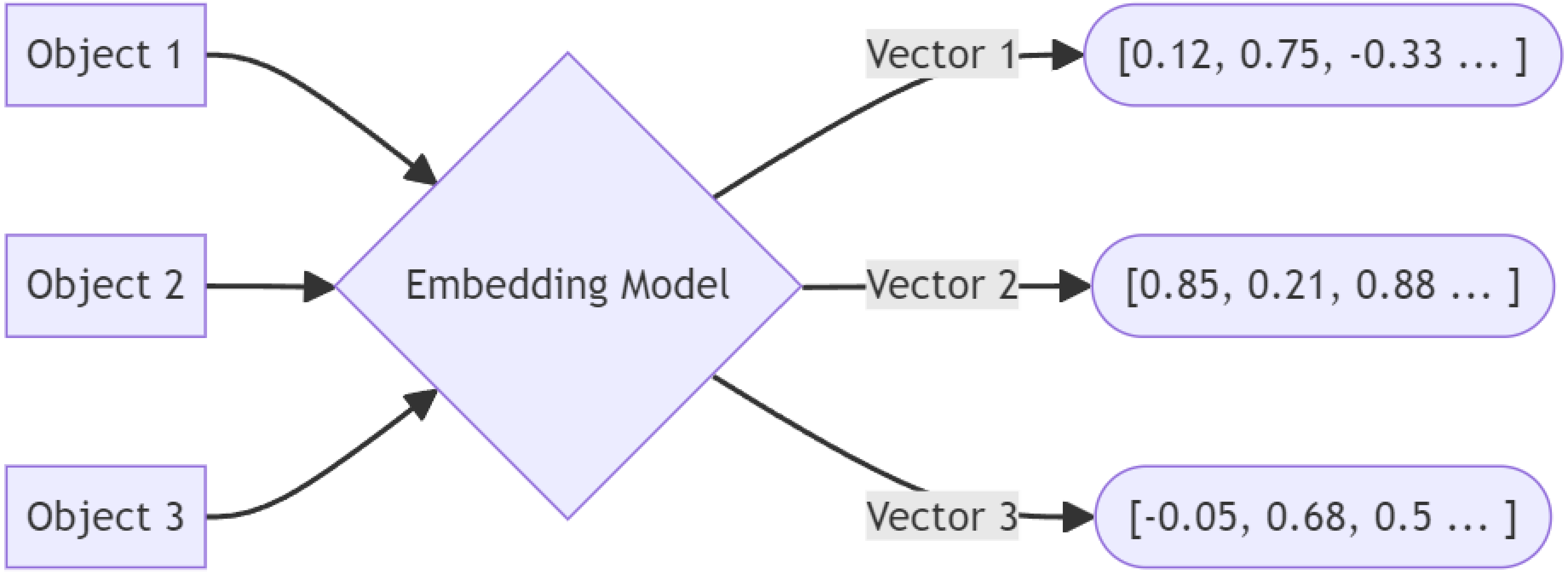
ACM CCS: • Human-centered computing → Visual analytics; • Human-centered computing → Information visualization; • Human-centered computing → Visualization systems and tools; • Computing methodologies → Machine learning; • Applied computing

1. Introduction

In recent years, the concept of *embedding technology* has gained a lot of attention within the research community. The term "embedding" refers to the process and/or the results of projecting data entities into a (relatively) low-dimensional space where the similarities of different points reflect their semantic closeness in the original space [MLMRC18, GWW19]. (Besides the process, the resulting *embedding vectors* are often referred to as "embeddings", too.) Various embedding approaches have been developed based on a target application domain and data types, such as numerical values [BCV13], word/text [MSC*13, LM14, AX19, FYC*22], graphs/networks [HYL17, GF18, ZYZZ20], media data [AGH*23], or a combination of those [BCV13, WMWG17]. Learning effective embeddings is crucial for performing downstream tasks in various fields, such as information retrieval [JSR*19], natural language processing [EAKC*20], social network analysis [AAM*21], and urban planning [MHL*20]. To further enhance the accuracy, explainability, and credibility of either the embedding process or the higher-level objectives, many studies incorporate a human-in-the-loop process in addition to optimizing the computational components [EHR*14]. There is a need to explore relationships within the embedding space and interpret the embedding process—and that is where visual analytics steps in.

In information visualization (InfoVis) and visual analytics (VA), many studies have been carried out on creating interactive user interfaces (UI) to understand embeddings over the past years. The rapid development of embedding methods, especially in deep representation learning [GWW19], enables many VA tools that acquire non-numerical data to perform new types of joint analysis, creating customized visual encodings and interactions. As Figure 1 shows, embeddings can be computed as part of VA pipelines or provided as part of input data. They are associated with various approaches for performing analytical tasks, encoding visual meanings, and enabling user interactions within the embedding space. These embeddings can be evaluated with regard to both their computational processes and their visual analytical aspects. The methodology and application scenarios for embedding technologies are highly heterogeneous. Furthermore, the terminology of the field is far from being standardized [KØSV18], making it hard to get a comprehensive overview for the respective interested readers.

What are embeddings?



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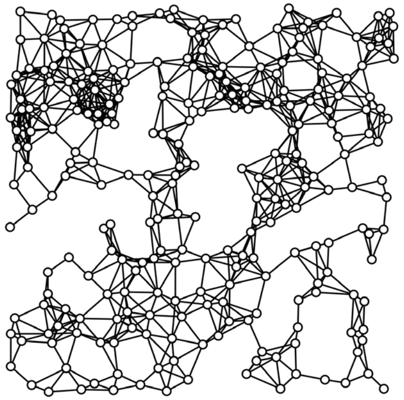
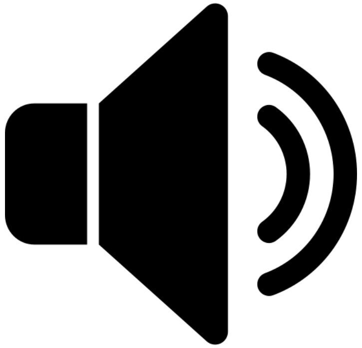


Embedding model

0.5	0.3	0.1	0.8	...	0.9	0.1

Embedding model

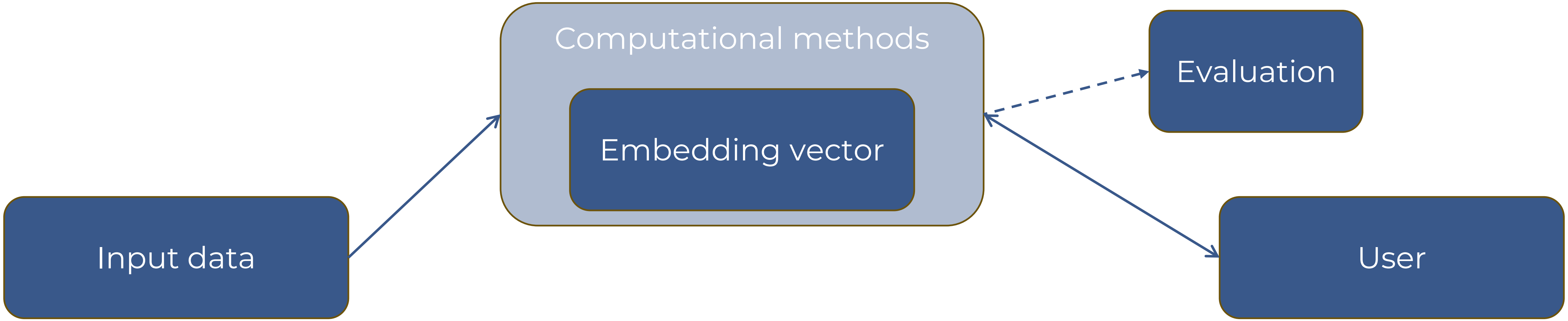
0.3	0.3	0.1	0.7	...	0.2	0.1



Embedding model

0.6	0.2	0.9	0.8	...	0.9	0.5

What are embeddings?



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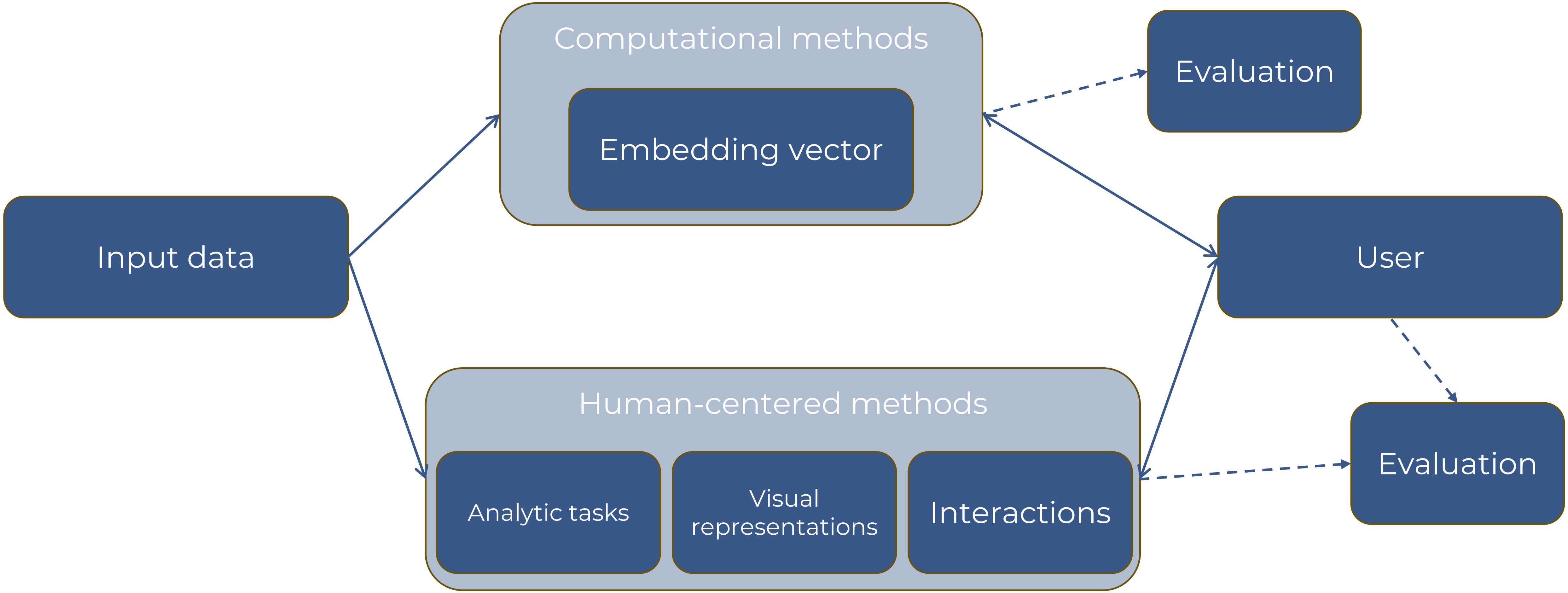

Human-in-the-loop processes can augment computer-centric components.

Applications of embeddings

Application Domain	122
ML, AI, Data Science	57
Computing	24
Life Sciences and Medicine	13
Physical Sciences, Engineering, Mathematics	4
Humanities, Social Sciences, Education	23
Business, Management, Governance, Law	12
Sports and Entertainment	3
Domain-Agnostic	12
Free-text details	*122*

Target User	32
Explicit Target User Description	32
Free-text details	*33*

Input Data Type	122
Numerical/Tabular	25
Textual	70
Graph/Network	22
Image/Video/Audio	17
Mixed	17
Data-Agnostic	6
Free-text details	*59*

Computational Method	116
Neural Network	88
Matrix Analysis	33
Statistical Analysis	16
Computational Method-Agnostic	12
Free-text details	*122*

Embedding Vector Dimensionality	122
Under 50 Dims.	15
Between 50–500 Dims.	31
Above 500 Dims.	14
Embedding Dims. N/A	70

Visual Analytic Task	118
Model Construction	14
Model Debugging/Quality/Bias Control	8
Model Results Representation	63
Model Results Explanation	23
Interactive Exploration	53
Comparison/Selection	52
Free-text details	*122*

Visualization Aspects	102
Explicit Embedding Representation	102
Free-text details	*107*

Interaction Aspects	77
Interaction Techniques Support	77
Free-text details	*79*

Evaluation Aspects	98
Evaluation of Computational Components	53
Evaluation of Visual/Interactive Components	84
Free-text details	*101*

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Applications of embeddings

https://va-embeddings-browser.ivis.itn.liu.se/

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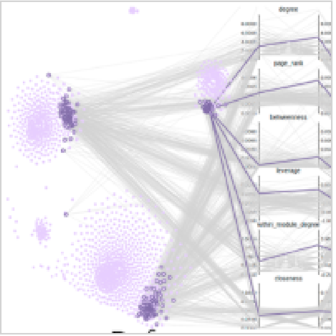
Applications of embeddings in visualization

- What is the data?
- What is the computational method?
- What is the visual analytics task?
- What is the visualization?
- What is the interaction?

Applications of embeddings



RoleSeer (2022)

by Xie, Laixin; Wu, Ziming; Xu, Peng; Li, Wei; Ma, Xiaojuan; Li, Quan

Application Domain/Task Details: gameplay, social network

Target User Details: game designers and game user experience (UX) practitioners

Input Data Details: friendship network

Computational Method Details: Diachronic Node Embedding, struc2vec, dynAERNN with encoder-decoder architecture

Visual Analytic Task Details: represent structural identity, exploring potential roles

Visualization Details: 2D projection, PCP

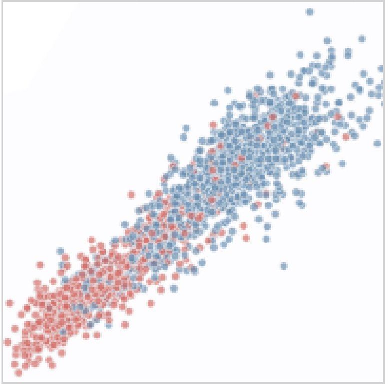
Interaction Details: click, lasso

Evaluation Details: effectiveness

Xie, Laixin, et al. "RoleSeer: Understanding informal social role changes in MMORPGs via visual analytics." CHI Conference on Human Factors in Computing Systems. 2022.

URL: https://doi.org/10.1145/3491102.3517712

BibTeX



DG-Viz (2020)

by Li, Rui; Yin, Changchang; Yang, Samuel; Qian, Buyue; Zhang, Ping

Application Domain/Task Details: electronic medical record analysis

Computational Method Details: RNN

Visual Analytic Task Details: analyze patient similarity

Visualization Details: 2D projection

Evaluation Details: baseline comparison, usefulness

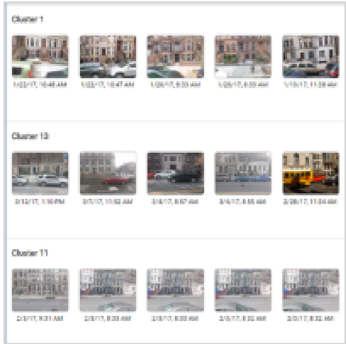
Li, Rui, et al. "Marrying medical domain knowledge with deep learning on electronic health records: A deep visual analytics approach." Journal of Medical Internet Research 22.9 (2020): e20645.

URL: https://doi.org/10.2196/20645

URL: https://github.com/DGViz/DGViz.github.io

BibTeX

Applications of embeddings



Urban Mosaic (2020)

by Miranda, Fabio; Hosseini, Maryam; Lage, Marcos; Doraiswamy, Harish; Dove, Graham; Silva, Cláudio T.

Application Domain/Task Details: urban planning, visual (image) search interface

Target User Details: practitioners in urban planning

Input Data Details: images

Computational Method Details: get intermediate layers of pre-trained CNN

Visual Analytic Task Details: image similarity search, represention of visual information of an image

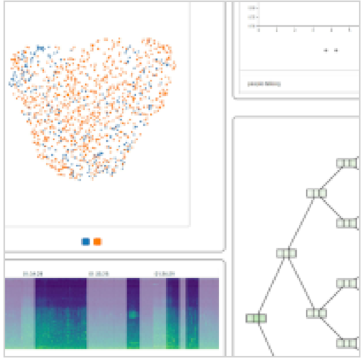
Visualization Details: compute similarity metrics to display clusters of similar images from queries

Evaluation Details: trading off accuracy for speed

Miranda, Fabio, et al. "Urban Mosaic: Visual exploration of streetscapes using large-scale image data." Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. 2020.

URL: hhttps://doi.org/10.1145/3313831.3376399

BibTeX



Urban Rhapsody (2022)

by Rulff, Joao; Miranda, Fabio; Hosseini, Maryam; Lage, Marcos; Cartwright, Mark; Dove, Graham; Bello, Juan; Silva, Cláudio T.

Application Domain/Task Details: search query, urban science

Input Data Details: hours of recorded audio

Computational Method Details: OpenL3, self-supervised learning

Visual Analytic Task Details: allows users to query using an audio sample, and drill down to days containing similar audios

Visualization Details: 2D projection, tree

Interaction Details: stack, click, select

Evaluation Details: user study not explicitly specified

Rulff, Joao, et al. "Urban Rhapsody: Large-scale exploration of urban soundscapes." Computer Graphics Forum. Vol. 41. No. 3. 2022

URL: https://doi.org/10.1111/cgf.14534

URL: https://github.com/VIDA-NYU/Urban-Rhapsody

BibTeX

Urban Mosaic: street-level images

- *" ... a tool for exploring the urban fabric through a spatially and temporally dense data set of 7.7 million street-level images from New York City, captured over the period of a year. Working in collaboration with professional practitioners, we use Urban Mosaic to investigate questions of accessibility and mobility, and preservation and retrofitting."*

Urban Mosaic: Visual Exploration of Streetscapes Using Large-Scale Image Data

Fabio Miranda[1], Maryam Hosseini[2], Marcos Lage[3], Harish Doraiswamy[1], Graham Dove[1], Cláudio T. Silva[1]

[1]New York University; [2]Rutgers University; [3]Universidade Federal Fluminense

[1]{fmiranda,harishd,grahamdove,csilva}@nyu.edu; [2]mary.hosseini@rutgers.edu; [3]mlage@ic.uff.br

ABSTRACT

Urban planning is increasingly data driven, yet the challenge of designing with data at a city scale and remaining sensitive to the impact at a human scale is as important today as it was for Jane Jacobs. We address this challenge with Urban Mosaic, a tool for exploring the urban fabric through a spatially and temporally dense data set of 7.7 million street-level images from New York City, captured over the period of a year. Working in collaboration with professional practitioners, we use Urban Mosaic to investigate questions of accessibility and mobility, and preservation and retrofitting. In doing so, we demonstrate how tools such as this might provide a bridge between the city and the street, by supporting activities such as visual comparison of geographically distant neighborhoods, and temporal analysis of unfolding urban development.

Author Keywords

Urban planning; Interactive visualization; Data analysis; Urban data

CCS Concepts

•**Human-centered computing → Human computer interaction (HCI);** *Visual analytics; Visualization toolkits;*

"A sense of place is built up, in the end, from many little things too, some so small people take them for granted, and yet the lack of them takes the flavor out of the city."

(Jane Jacobs, Downtown is for People)

INTRODUCTION

For those of us living in or visiting the world's major cities, their dynamism and complexity are immediately apparent. Yet urban planners and designers must work in a context where any single intervention, perhaps aimed at altering just one aspect, can have a wide ranging impact on a variety of interrelated components [12], affecting things at both a macro *city scale* and a micro *human scale*. Examples of changes at a city scale include, suburbanization, economic deconcentration, modification to transport infrastructure, rezoning and/or gentrification of neighborhoods, and major renewal projects. Examples of changes at a human scale, on the other hand, are reflected in the city's urban fabric, and include aspects that make a city livable, encourage walking, and contribute to the perception of safety; in other words affect the day to day lives of its inhabitants. This might be manifested in lighting, shadow, sky exposure, open-front shops, the details of building facades, etc. [32].

Responding to these challenges at the city scale, planners and designers have been aided by a rapid growth in data from urban environments, and so are able to turn to computational methods and large-scale data analysis, which increase understanding by quantifying different aspects of the city (e.g., [36, 13, 14, 29, 70, 67, 48]). However, while sensitivity to the impact of change at a human scale remains as important today as it was for Jane Jacobs and others in the 1950s and 1960s [47], analyses of suitable data, which emphasize qualitative, visual details, are often difficult and time-consuming to perform. Although there is an increasing availability of street-view images, which support a degree of virtual assessment and auditing of the built environment, their distribution is often temporally sparse and so analysis is limited.

This paper introduces Urban Mosaic, a tool for visually exploring the urban fabric. It responds to the challenges practitioners face by employing a newly available spatially and temporally dense data set of street-level images from New York City (NYC) Urban Mosaic is a visual exploration system designed to help practitioners in urban planning and design gain insight into the human scale impact of changes in the urban fabric. It utilizes state-of-the-art computer vision techniques for image similarity search and clustering, together with efficient spatio-temporal selection and aggregation over the image metadata to visually explore and map this image data set. It further allows the analysis to be augmented using spatio-temporal urban data from a variety of other sources (e.g., census, transport, crime, weather, housing market, zoning, noise complaints). The image data set used in this work contains 7.7 million images captured in the Manhattan and Brooklyn boroughs between April 2016 and April 2017 using car-mounted cameras. Urban Mosaic has been developed as a collaboration between researchers in urban planning, visual analytics, and HCI. We include a detailed

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http://dx.doi.org/10.1145/3313831.3376399

[Miranda et al., 2020]

Urban Mosaic: street-level images

- Demonstrates the potential for images from large-scale street-view data sets to help bridge data-driven urban planning and design at city scale and at the human level.
- Exploration of the urban fabric using large spatiotemporal collection of images.
- Visual comparison of geographically distant areas to analyze unfolding urban developments.

Urban Mosaic: Visual Exploration of Streetscapes Using Large-Scale Image Data

Fabio Miranda[1], **Maryam Hosseini**[2], **Marcos Lage**[3], **Harish Doraiswamy**[1], **Graham Dove**[1], **Cláudio T. Silva**[1]
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[Miranda et al., 2020]

Temporally-dense street-level images

Objectives

Support the interactive analysis of the city at the micro scale, over geographically distant regions.

Comparison of the urban fabric in different regions

Assessment of features in the built environment

Assessment of walkability and accessibility

Image query composition

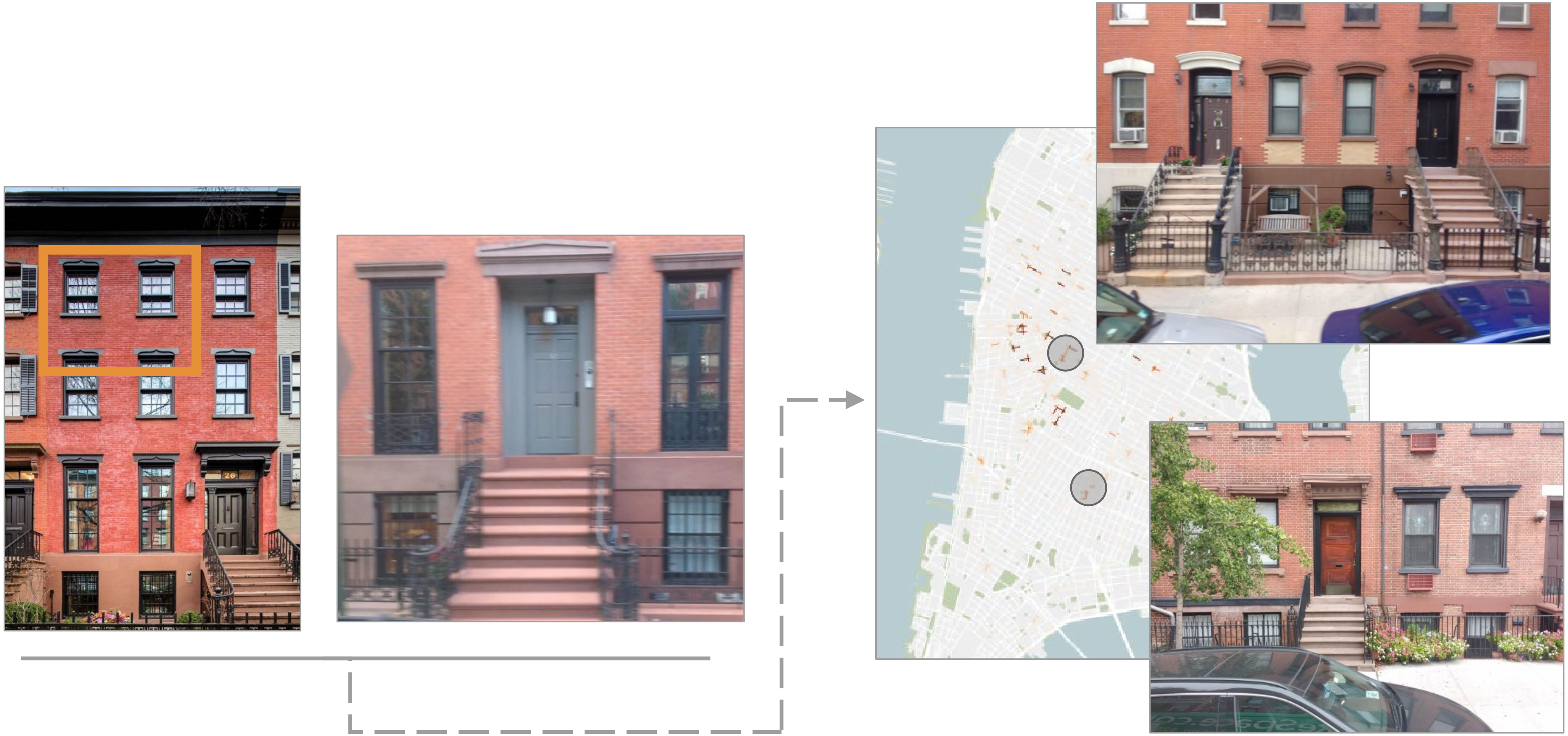


Image embeddings

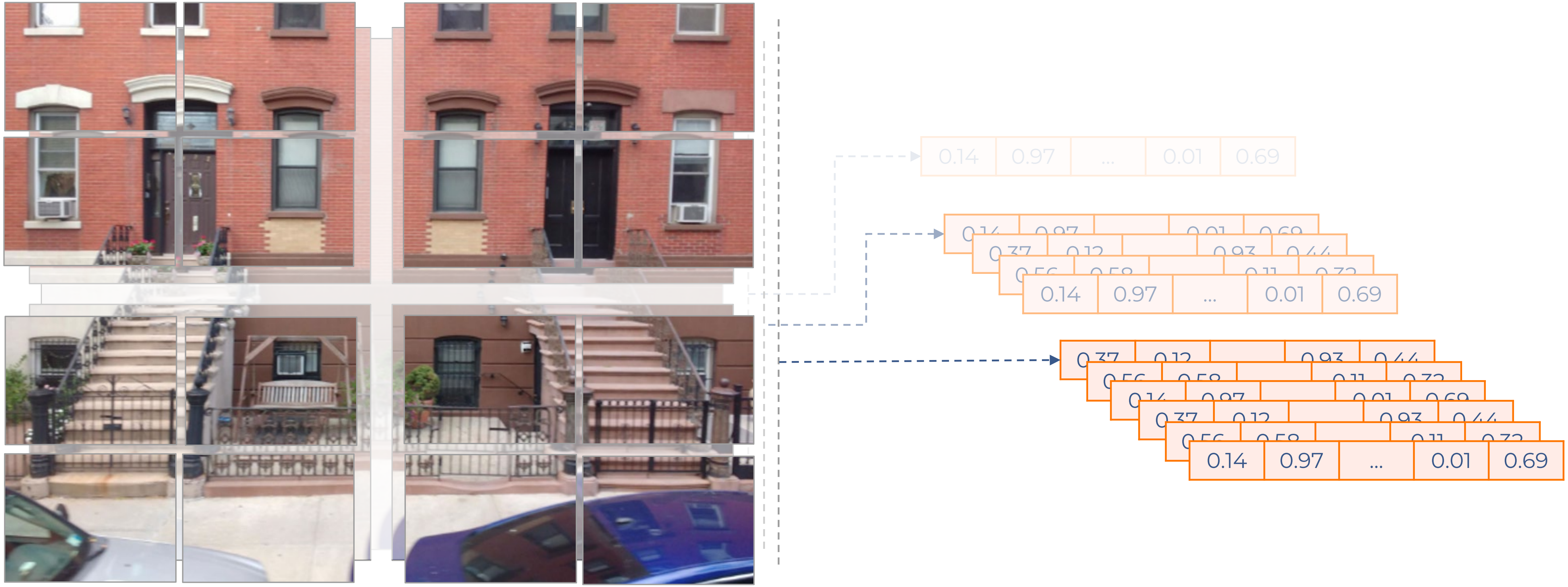


Image embeddings

```
# pip install img2vec-pytorch pillow scikit-learn
from img2vec_pytorch import Img2Vec
from PIL import Image

img2vec = Img2Vec(model='resnet18’) # produces a pooled
vec512 = img2vec.get_vec(Image.open("img.jpg").convert("RGB"))
```

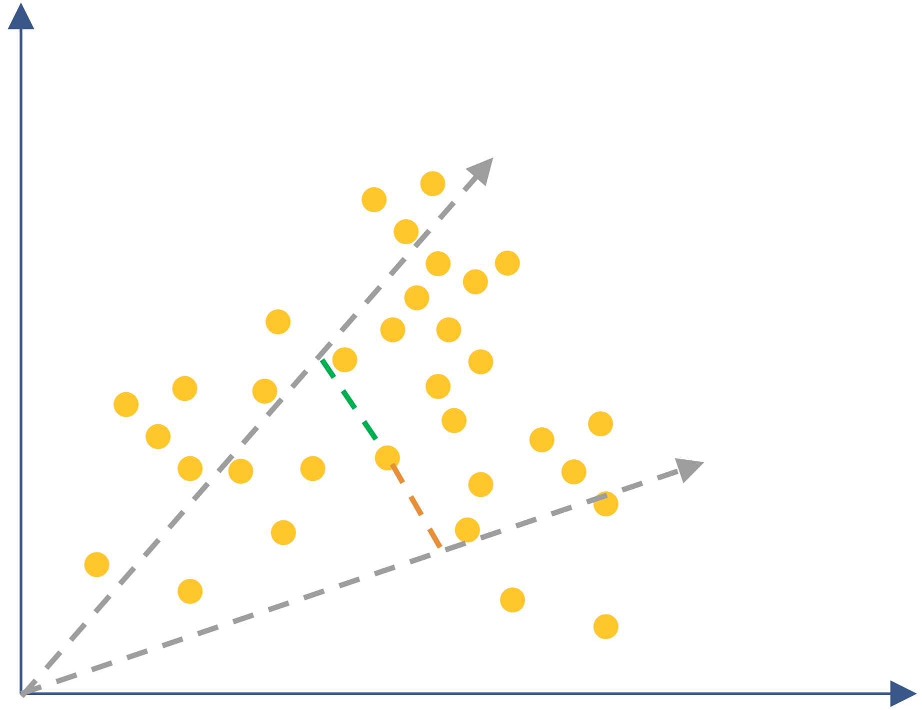
Image embeddings

```
# pip install sentence-transformers pillow scikit-learn
from sentence_transformers import SentenceTransformer
from PIL import Image

m = SentenceTransformer("clip-ViT-B-32")  # 512-D
img = Image.open("img.jpg").convert("RGB")
vec512 = m.encode(img, convert_to_numpy=True, normalize_embeddings=True)
```

Locality sensitive hashing

21 GB worth of embeddings data



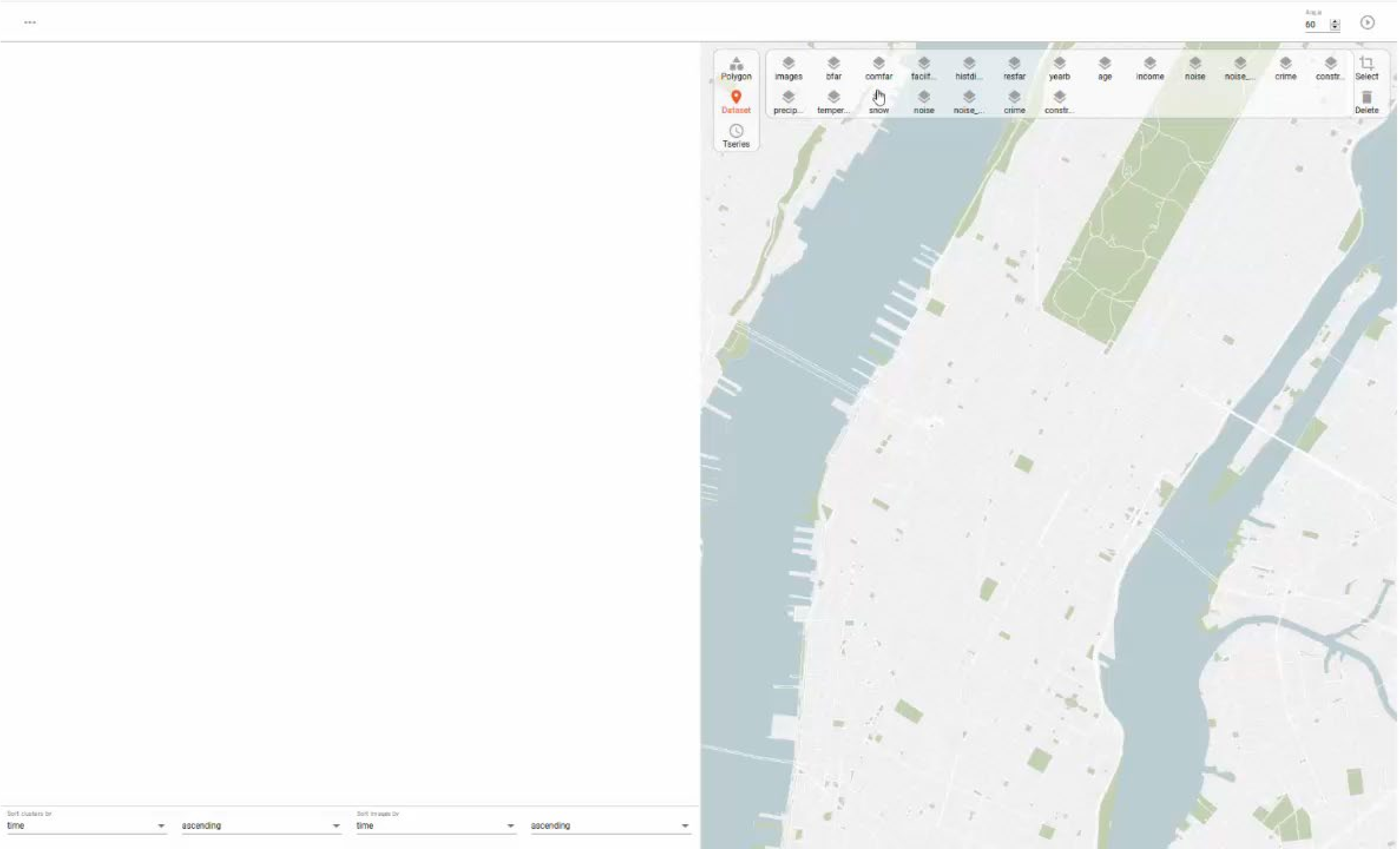
0.37	0.12	...	0.93	0.44

4 kbytes per embedding

0	0	...	1	0

64 bytes per embedding

$$\alpha_{1,2} = \cos^{-1}\left(\frac{\overrightarrow{v_1} \cdot \overrightarrow{v_2}}{|\overrightarrow{v_1}||\overrightarrow{v_2}|}\right)$$

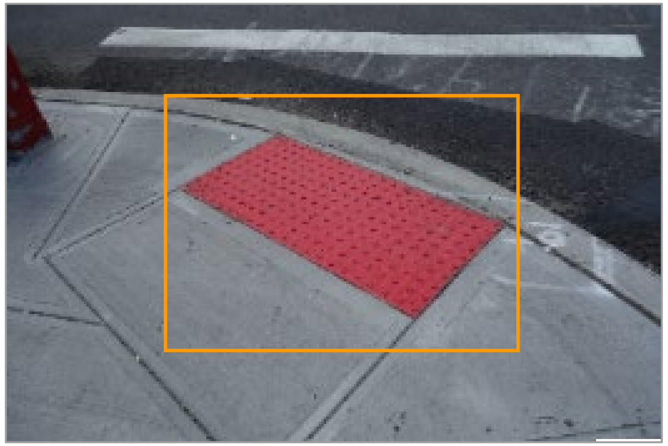

60
Polygon
Dataset
Tseries
images
bfar
comfar
facil...
histdi...
resfar
yearb
age
income
noise
noise_...
crime
constr...
precip...
temper...
snow
noise
noise_...
crime
constr...
Select
Delete
Sort clusters by
time
ascending
Sort images by
time
ascending

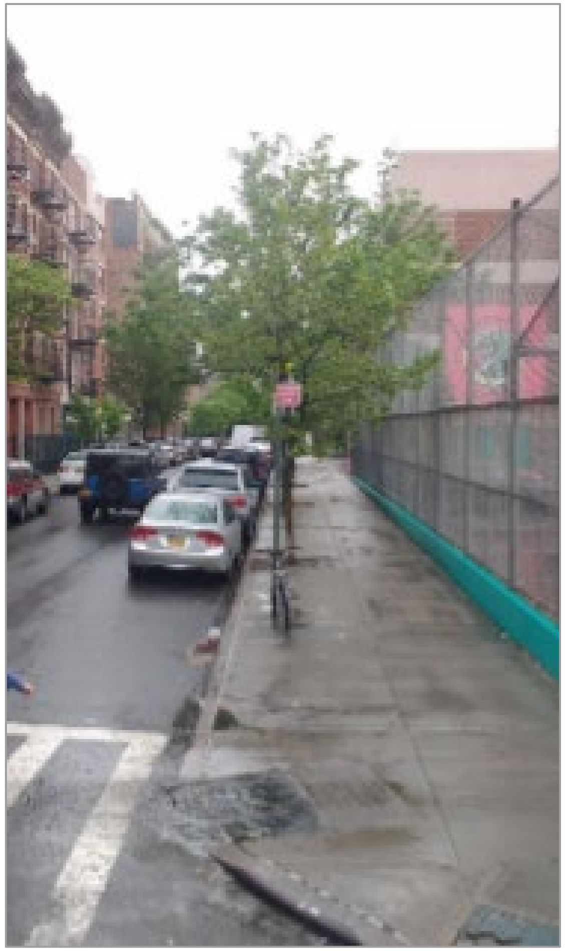
Accessibility for older adults

Focuses on interventions that enable older adults to “Age in Place” and prevent outdoor fall.

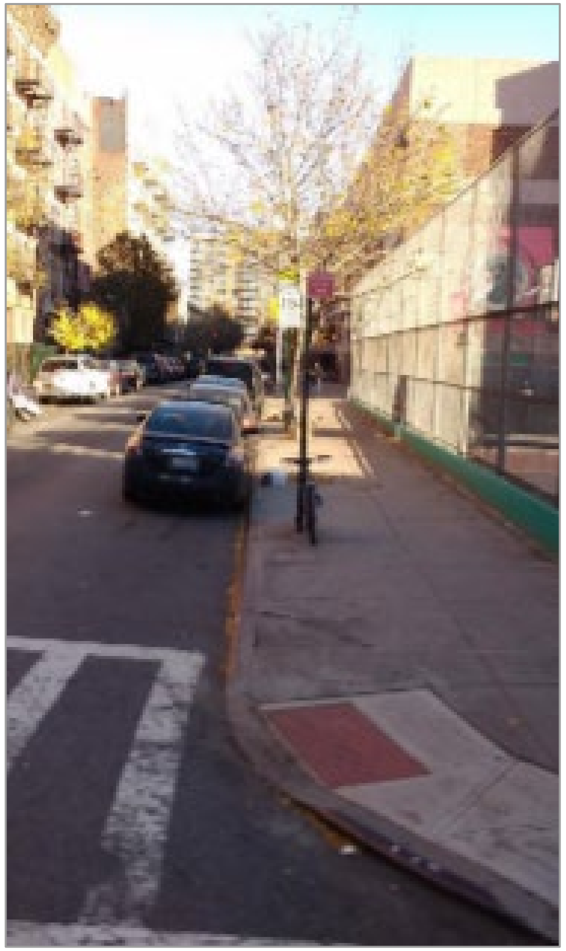
Impact of the neighborhood environment on falls.

Accessibility: installation of tactile pavings

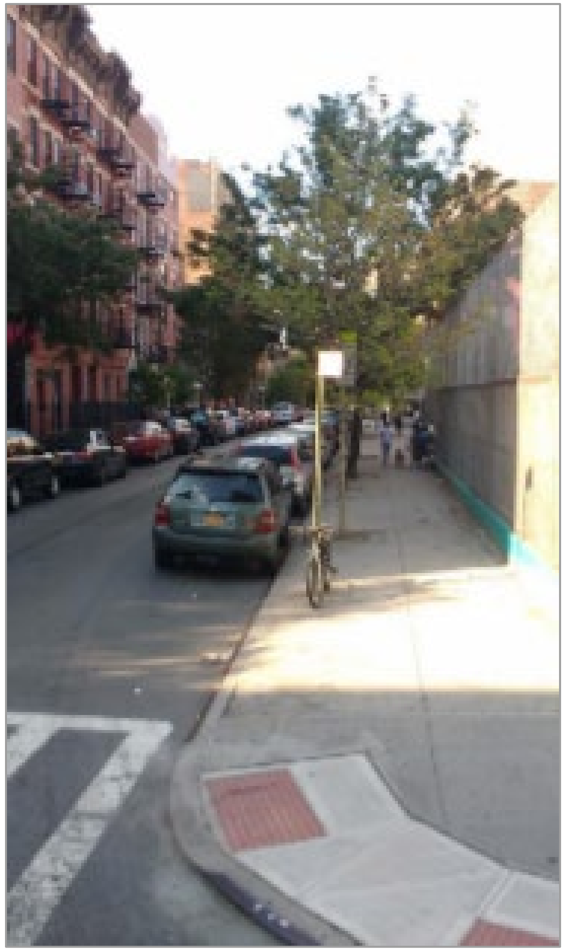




June

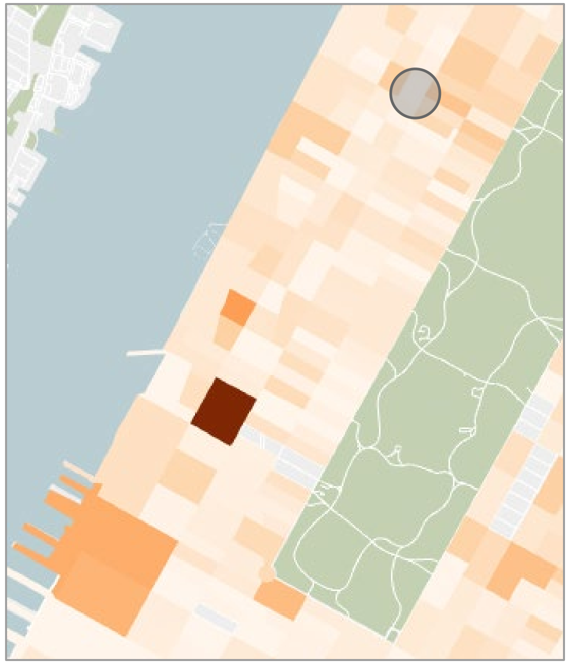


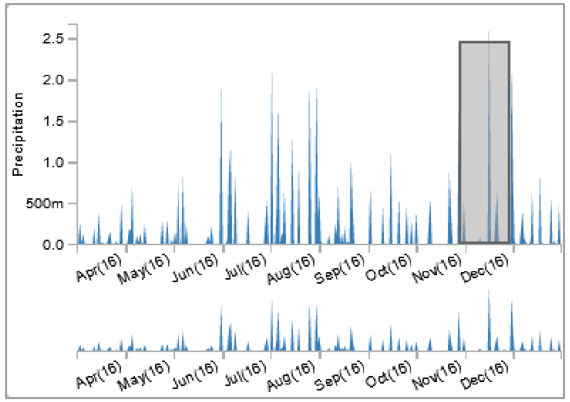
August

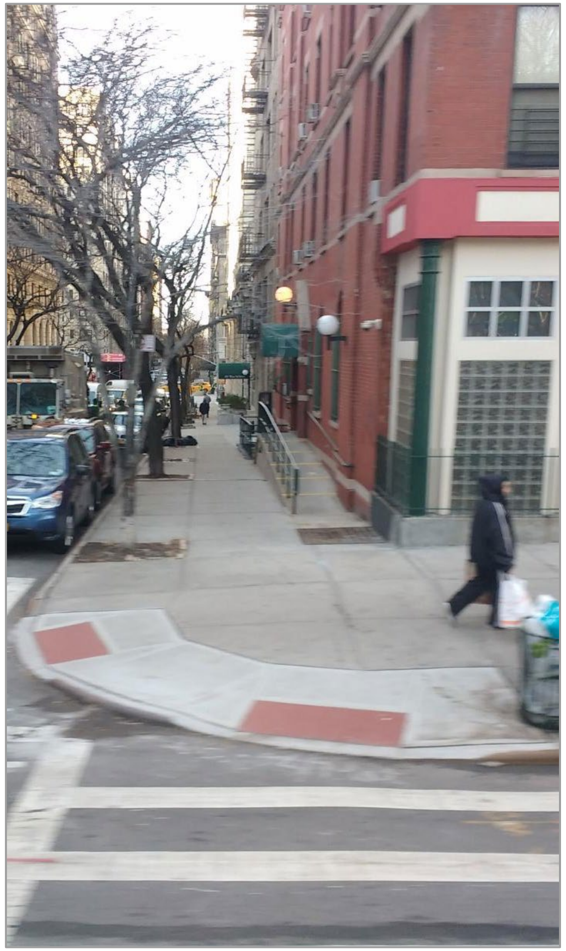


October

Accessibility: assessing hazards for older adults







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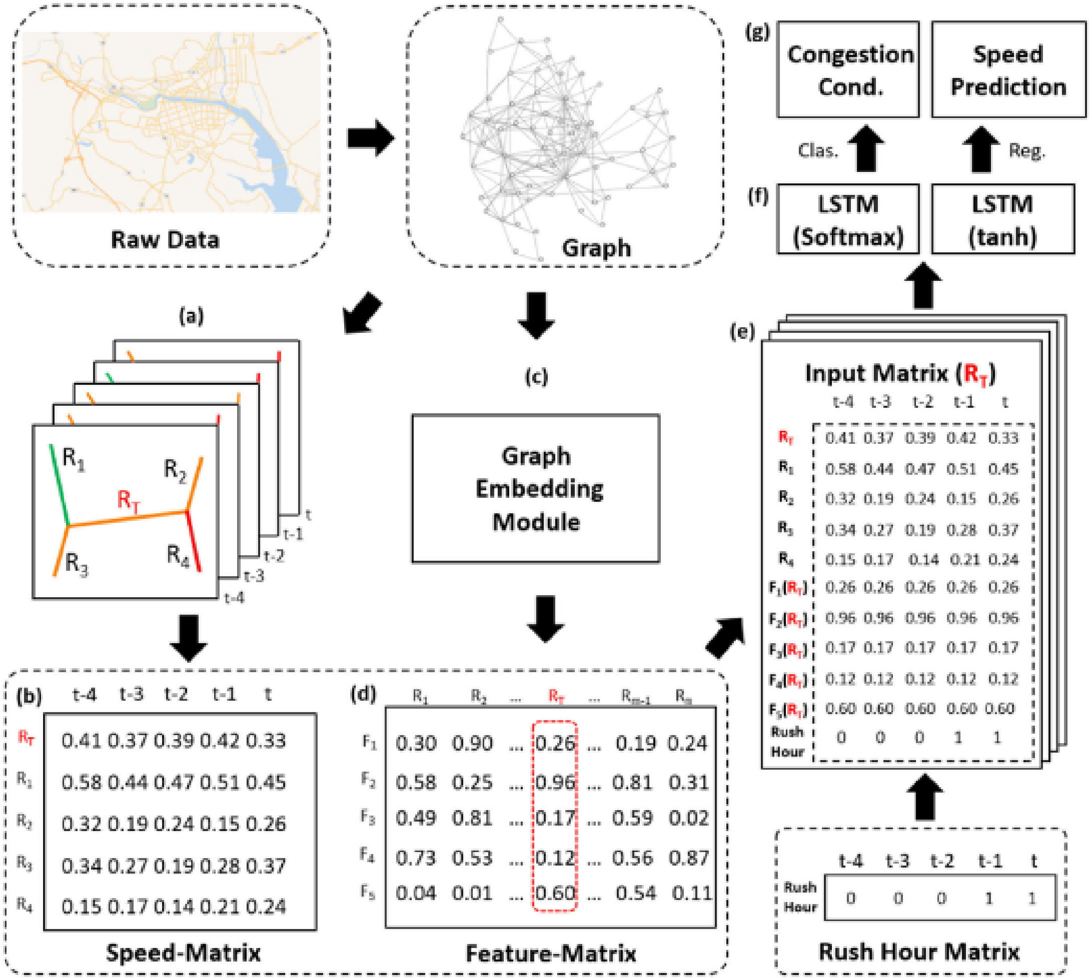
Main takeaways

- General approaches might not scale.
- Data visual analytics might require structures specifically designed for a given task.
- Understand the data, and how it can be transformed to best fit different computational resources and architectures.

Applications of embeddings in visualization

- What is the data?
- What is the computational method?
- What is the visual analytics task?
  - Model construction
  - Model debugging
  - Model results representation
  - Model results explanation
  - Interactive exploration
  - Comparison / selection
- What is the visualization?
- What is the interaction?

Embeddings for model construction



IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 26, NO. 11, NOVEMBER 2020 3133

A Visual Analytics System for Exploring, Monitoring, and Forecasting Road Traffic Congestion

Chunggi Lee, Yeonjun Kim, Seungmin Jin, Dongmin Kim, Ross Maciejewski, *Senior Member, IEEE*, David Ebert, *Fellow, IEEE*, and Sungahn Ko, *Member, IEEE*

Abstract—We present an interactive visual analytics system that enables traffic congestion exploration, surveillance, and forecasting based on vehicle detector data. Through domain expert collaboration, we have extracted task requirements, incorporated the Long Short-Term Memory (LSTM) model for congestion forecasting, and designed a weighting method for detecting the causes of congestion and congestion propagation directions. Our visual analytics system is designed to enable users to explore congestion causes, directions, and severity. Congestion conditions of a city are visualized using a Volume-Speed Rivers (VSRivers) visualization that simultaneously presents traffic volumes and speeds. To evaluate our system, we report performance comparison results, wherein our model is more accurate than other forecasting algorithms. We demonstrate the usefulness of our system in the traffic management and congestion broadcasting domains through three case studies and domain expert feedback.

Index Terms—Traffic, road, congestion, visualization, deep learning, LSTM, surveillance, forecasting, predictive analysis

1 Introduction

CONGESTION is one of urban transportation's most prevalent challenges, and identifying the true cause of congestion is a major strategic issue in urban planning. As such, many government organizations have invested considerable resources towards the installation of vehicle detectors and Intelligent Transportation Systems (ITS) [1] to analyze and monitor road congestion. However, analyzing and surveilling congestion is difficult due to the high volume and complexity of the data. This issue is further compounded by the lack of effective analytical systems. Analysts in a city traffic management center use vehicle detector data for congestion analysis, but their analysis is mainly performed with table-based statistical tools and simple visualizations (e.g., EXCEL, SPSS). Similarly, reporters in a radio station report on congestion information (e.g., traffic accidents) that has been collected by monitoring hundreds of CCTVs. However, monitoring a vast number of surveillance cameras is difficult, and important information that could affect near-future traffic conditions (e.g., illegal parking, vehicle accidents) is often missed. As such, there is a need for systems that can monitor a variety of streaming data from multiple sources and integrate this information for real-time congestion monitoring, prediction, and retrospective analysis.

Many visualization systems for traffic analysis have been proposed, but they cannot be easily adapted for large cities' real-time traffic surveillance, analysis, and management. Two systems exist for traffic surveillance [2], [3], but they are not scalable for large city traffic monitoring and analysis tasks, as their focus is on small area surveillance. Other traffic analysis systems [4], [5], [6] can be used for analyzing wider areas, but their traffic data are calculated from GPS trajectories, which is not preferred for real-time surveillance due to computation and time costs. Above all, no existing visual systems provide congestion forecasts in real time, which is important for many tasks such as traffic signal adjustment and radio broadcasting traffic conditions.

In this paper, we present a visual analytics system to support analysts and reporters in the congestion management and traffic broadcast domains. We have conducted interviews with analysts in these domains to derive task requirements and found that the analysts want to see traffic patterns (flow and velocity) and congestion. For this, we provide two views (StreamView and PropagationView) to view the congestion in different perspectives. In the StreamView, we provide Volume-Speed Rivers (VSRivers) that show an entire city's congestion conditions on a map by representing roads' traffic volumes and speeds with shapes and colors. In the PropagationView, we provide Propagation Arrows (PropArrows), a node-link diagram application, to show how congestion propagates on the map with arrow tails and heads representing congestion start and end locations respectively. VSRivers and PropArrows form the basis of our multiple-coordinated views [7] and allow users to 1) explore temporal congestion patterns with estimated congestion propagation, 2) monitor real-time congestion conditions,

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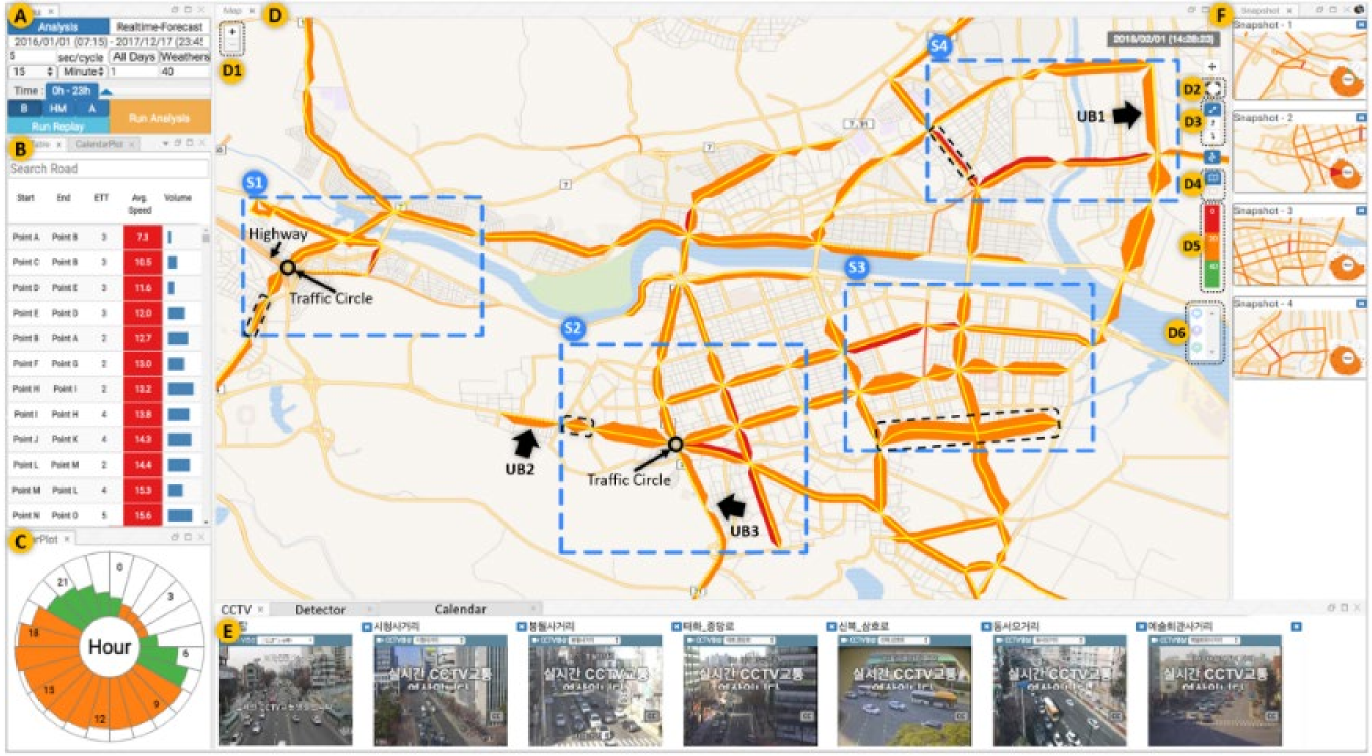
Embeddings for model construction



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